

The Half-Life of Compute

What export controls on AI chips actually buy, and for how long

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Every argument for restricting the export of advanced AI chips rests on a claim about time: controls **buy time**. Nobody has ever said how much. This paper computes it, and finds that the usual explanation of why controls work is the wrong one.

The mistake: treating chips like gold

National AI capacity is measured constantly. At least eleven public indices rank countries by how much computing power, talent and infrastructure they hold. All of them take a photograph. None asks what happens to the picture afterwards.

That matters, because installed computing power is not a reserve sitting in a vault. It is a fleet of machinery that breaks. Using published failure data from a real training run, roughly **9% of accelerators fail in a year** from causes that cannot be repaired in the field. They also age against a target that keeps moving: the scale of a state-of-the-art training run roughly doubles every ten months. A gold reserve does not corrode. A chip fleet does, twice over.

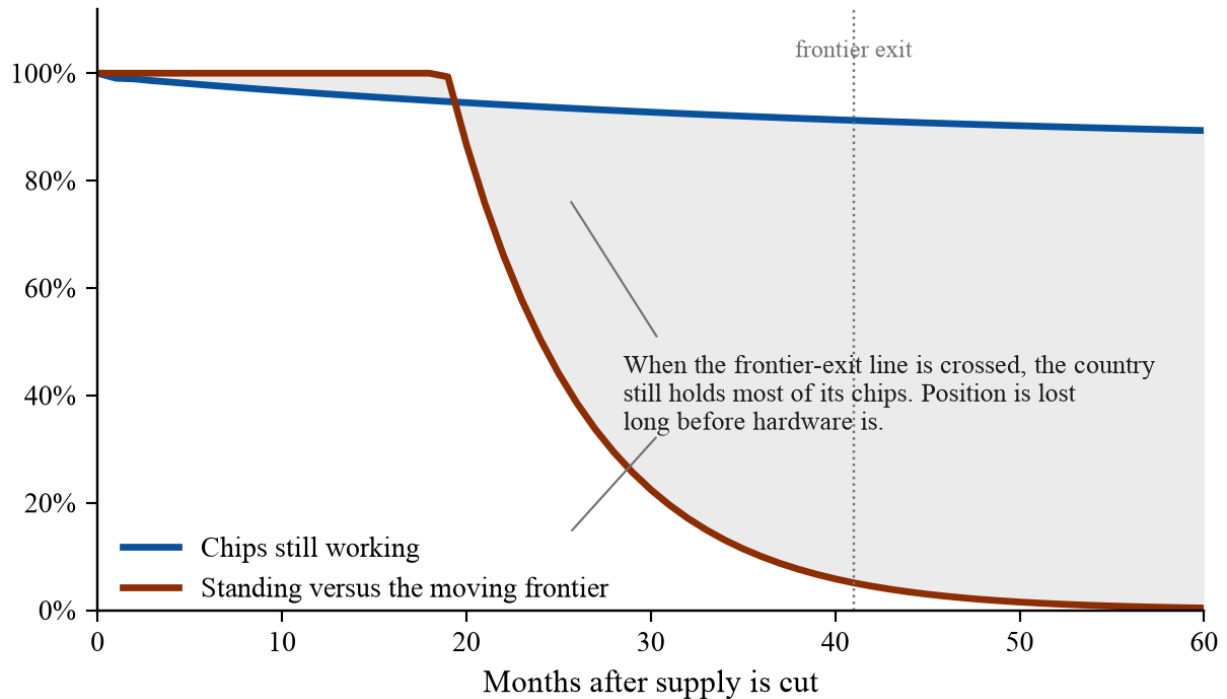
The engineering literature has measured the corrosion precisely, but always assuming replacement chips can be bought. That assumption is exactly what a supply cut-off removes. Removing it is what this work does.

Three findings

1 Two to four years, and stockpiling barely extends it

A cut-off country falls below a competitive share of the frontier in **25-49 months**. The interval grows only with the *logarithm* of how many chips you hold: **ten times the compute buys about seventeen extra months**, a hundred times about thirty-five. And hardware failure accounts for only 13-14% of that clock, the rest is the frontier moving away from you.

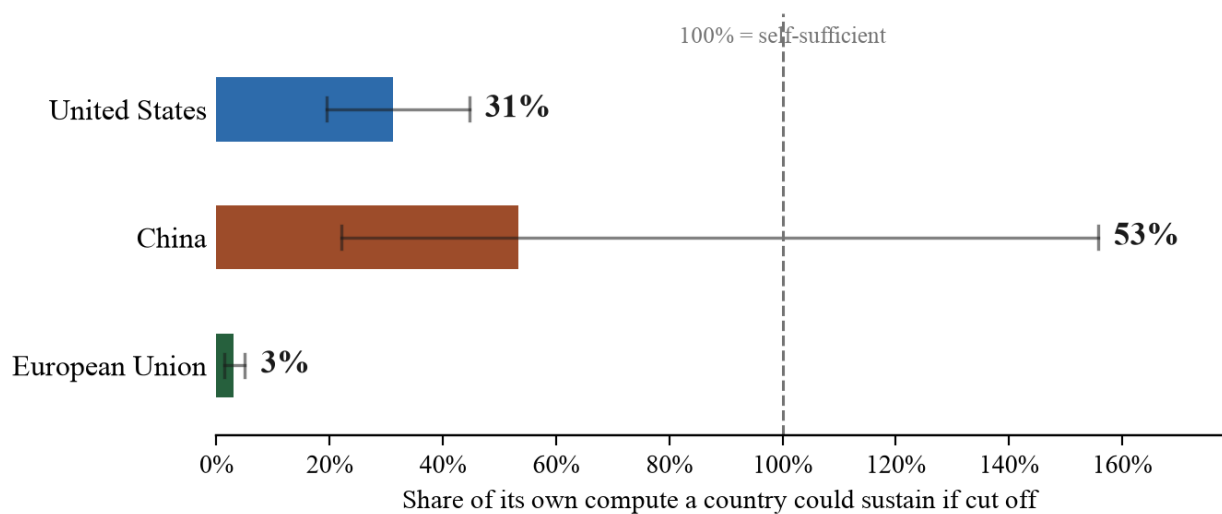
Your hardware survives. Your position does not.



The gap between the two lines is the finding. Hardware declines slowly; competitive standing collapses quickly, because the frontier keeps moving away. An earlier draft of this work claimed the interval did not depend on stock size at all, that turned out to be an artefact of how the measure was built, and has been withdrawn.

2 Nobody can show self-sufficiency, and for China it is genuinely undecided

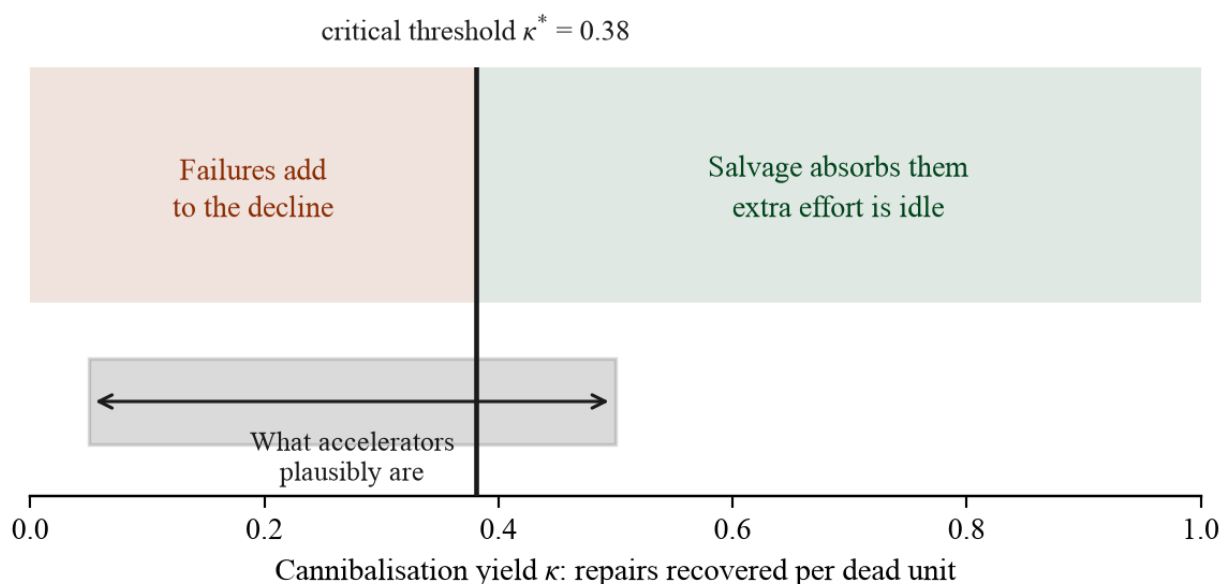
The **sovereign floor** is the share of its own computing power a country could sustain from non-severed inflow if cut off. At the median it is **31.1%** for the United States, **53.4%** for China and **3.1%** for the European Union, but the three numbers mean different things. The US and EU figures are largely assumptions about domestic production, rescaled. China's is a genuine model output, and its uncertainty range runs from 22.1% to 155.7%, crossing full self-sufficiency at the top. Current public data cannot settle whether China could sustain its fleet alone. Lithography is Dutch, leading-edge fabrication is Taiwanese, the specialised memory is Korean: nobody exits this system without cost.



Bars are median estimates; the thin lines show the 90% uncertainty range. Important caveat: for the United States and Europe this number is largely an assumption about how much they could make domestically, rescaled, only China's is a genuine model output. See the caveats overleaf.

3 A hidden switch nobody has measured

When resupply stops, the only source of spare parts is other machines. In aviation under sanctions, operators strip roughly one airframe to keep ten flying. AI accelerators cannot be salvaged that way: the memory is bonded into the same package as the processor, so the two most common failures are unreparable by design. There is a critical ratio, call it **0.38**, below which broken machines pile up and add to the decline, and above which salvage absorbs them completely and any further repair effort is wasted. The plausible range for real accelerators sits **across** that line. Below it a fleet contracts up to 40% faster than ageing alone would cause; above it, nothing more can be squeezed out. No public evidence yet says which side real hardware is on.



Establishing this single number empirically would be, on this analysis, the most valuable measurement the field could make.

What it means

Controls do not need to destroy an adversary's computing power. They need only freeze it while the frontier moves.

This is a different instrument from the one usually described, and it has different properties. Sensitivity testing shows the pace of frontier advance matters about **five times more** than hardware reliability (15 months of swing versus 2). The constraint that binds is the clock, not the machinery.

- **Controls are a bet on continued exponential progress.** If frontier advance slows, through scaling limits, energy constraints or capital withdrawal, their value decays with it. They are most effective exactly when they are least needed, and weakest when progress stalls.
- **Stockpiling buys less than it looks like.** A large holding of chips does not extend the horizon much, because the horizon is set by how fast the frontier moves, not by how much hardware you have.
- **Sovereignty needs a number, not an adjective.** The sovereign floor gives one. On these estimates no major power exceeds roughly a third.

One further result speaks directly to enforcement. Controls leak, and sweeping the leakage rate separates two things usually discussed as one. Smuggling at roughly the historically estimated rate lifts a country's sovereign floor from 64% to 100%, but buys only about 2 extra months at the frontier. To double the frontier horizon you would need leakage of roughly 200% of the installed base every year, which is not smuggling but an open market. **Leakage buys capacity, not competitiveness**, so interdiction protects a frontier lead far better than it

prevents an adversary holding a floor.

What we do not know

Three limits are worth stating plainly, because the numbers above will otherwise be read as more precise than they are.

- **The salvage ratio has never been measured.** It is bounded here by reasoning about how the chips are physically built, not by observation, and it straddles the critical threshold.
- **Chip ownership data resolve by company, not by country.** Assigning a multinational's fleet to a jurisdiction requires assumptions. Ranges are swept rather than asserted; European figures are the weakest.
- **No country has yet been fully cut off from AI compute.** We attempted to calibrate against the closest real case, Russian civil aviation after February 2022, and *the calibration failed*. Much of that fleet's decline came from aircraft being repossessed by their leasing owners and grounded by regulators, neither of which has any equivalent for computer hardware. We report the failure rather than force a fit, and it marks out what a real test would require.

How to use this

The full paper gives the model, the data sources for every parameter, and runnable code with a fixed random seed, so any figure here can be reproduced or re-run under different assumptions. Three quantities are meant to be reused directly: the **compute half-life**, the **frontier exit time**, and the **sovereign floor**. Available in English, Italian and French; the English version is the version of record.

Summary of a working paper. Full methods, data and code in the accompanying paper. DOI 10.5281/zenodo.21866487. Code: github.com/diShine-digital-agency/The-Half-Life-of-Compute